Discover: Enzyme Action: Preparing the pH Investigation

Today I can… □ Use these words in the right way. Use them to explain an idea: enzyme, substrate, active site and specificity. □ Say how an unsuitable pH can change the shape of an active site. Say how this could slow the enzyme's work. □ Name the things in the test: pH is what we change. It is the independent variable. Time until starch disappears is what we measure. It is the dependent variable. Choose suitable control variables. These are things to keep the same.

Feedback: □ VF □ WS □ I □ NS □ E □ R □ [//] □ [?] _____

Teacher and cover plan

1. Lesson outcome SCI-LAUNCH-A1-W08-L1 · Science · Autumn 1 · Week 8 I can explain enzyme specificity and denaturation, then identify the variables and evidence needed to investigate the effect of pH on amylase safely.

2. Arrival 1–4 Four questions; see STAFF ANSWERS · Arrival and the matching tier sheet.

3. Starter Which iodine observation shows starch is still present: blue-black or orange-brown?

4. I Do I do · bind, break down, release; I do · connect pH to the measure. In your book: Watch the model. No writing yet.

5. We Do We do · label the enzyme model; We do · organise the investigation. In your book: Record one evidence claim and explain its limit.

6. You Do Complete the chosen route using the shared evidence. In your book: Improve one explanation; mark the change // and read it back.

7. Support routes Independent → Supported → Choice; Aspire separately. Staff mark I, S or C beside this task.

8. Adaptations / normal ways of working Use planned adaptations in TA brief. Private response, seated or adult-directed access; agree a pause and return step.

9. TA focus No cold calling; nobody is asked to read difficult text aloud. Keep the first response. Use the ladder, then fade support.

10. Exit Two checks; use the matching tier. Staff guidance: STAFF ANSWERS · Exit.

11. After teaching Misconception: _____ Response: _____ Next step: _____

12. Preparation / evidence / cover essentials Use supplied models, cards, data and book. Read TA brief: Preparation and practical safety before teaching. Staff answers: Arrival, Exit, shared tasks and mark scheme. Use the supplied models, cards, data and booklet. This lesson plans only. During the later practical, handle reagents only with the centre-approved method, prepared equipment, local risk assessment and trained supervision. The supplied-observation route needs no reagents and teaches the same interpretation; it does not certify physical practical completion. Do not infer concentrations, heating arrangements or disposal from lesson cards. Further safety and preparation: continued on page 2.

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Teacher and cover plan · timeline and support

Prompt ladder for adults 1. Point to the relevant place / instruction. 2. Ask one short question: "What is it asking you to find?" 3. Rephrase the instruction in simpler language. 4. Provide a scaffold or limited choice. 5. Model the first step only, then return the task to the pupil.
00–04 · Arrival task · 4 min TA: Help locate and begin. Do not answer the questions.
04–08 · Starter · what can iodine tell us? · 4 min TA: Help access the question. Do not provide the response.
08–13 · I do · bind, break down, release · 5 min TA: Support attention. Do not give a parallel explanation.
13–18 · We do · label the enzyme model · 5 min TA: Prompt contribution and recording. Do not complete the guided response.
18–22 · I do · connect pH to the measure · 4 min TA: Support attention. Do not give a parallel explanation.
22–26 · We do · organise the investigation · 4 min TA: Prompt contribution and recording. Do not complete the guided response.
26–34 · You do · 8 min · KEEP TA: Use the ladder; record help and return to independence. Do not give the content answer.
34–37 · Review and improve · 3 min · KEEP TA: Receive the pupil-owned change. Do not change the pupil's meaning.
37–40 · Exit ticket · 3 min · KEEP TA: Ensure an attempt. Do not change an answer.
KEEP / shorten Keep independent evidence, review and exit. Take one shared explanation if short of time.
Evidence adult Teacher or TA receives the explanation; record actual support. Keep raw observations distinct from illustrative data. Use the lesson code on the evidence app.
Preparation and safety · continued Choose a route at each task using what the pupil shows. Switching adds no task, record or label. Use paper models or teacher-authorised apparatus. No tasting, personal genetic disclosure, physical contact or unplanned chemical handling. Offer adult-directed handling and a seated route.